

DIGITAL SIGNAL PROCESSING LAB REPORT

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Overview

This report summarizes the results of the Digital Signal Processing laboratory exercises, specifically focusing on generating, manipulating, and analyzing discrete-time signals and systems using Scilab. The report covers Lab 1 and Lab 2.

1 Introduction

The laboratory exercises provide practical experience with continuous-time to discrete-time signal conversion, signal properties, and elementary discrete-time operations. By utilizing Scilab, theoretical concepts such as sampling, quantization, time-shifting, folding, and signal arithmetic are modeled and visualized to deepen our mathematical understanding.

2 Objectives

The objectives achieved upon completing these laboratory sessions include:

- Utilizing Scilab for vector and matrix initializations.
- Understanding the sampling theorem and the quantization process (ADC procedure).
- Visualizing discrete signals via functions such as `plot2d3` and utilizing graph annotations.
- Implementing fundamental operations on discrete-time signals, including addition, multiplication, time-shifting, and time-reversal.

- Extracting the even and odd components of a given discrete-time sequence.

3 Methodology

The exercises were conducted on a Windows environment using the Scilab numerical computing software. Mathematical formulas from the time-domain representation of discrete signals were translated into Scilab scripts. Vectorized operations and padding techniques were applied where necessary to align discrete sequences prior to manipulation.

4 Practical Contents

4.1 Lab 1

Exercise 1.1

Question: Using the operators in vector and matrix to do the following tasks:

- Create a vector in the form of $(x_1 + 1, x_2 + 1, x_3 + 1, x_4 + 1)$ where x_1, x_2, x_3, x_4 are components of vector $x = 1 : 4$.
- Create a vector in the form of $(x_1y_1, x_2y_2, x_3y_3, x_4y_4)$ where $x_1 \dots x_4$ and $y_1 \dots y_4$ are components of vector $x = 1 : 4$ and vector $y = 5 : 8$, respectively.
- Create a vector in the form of $(\sin(x_1), \sin(x_2), \dots, \sin(x_{10}))$ where x is a vector of 10 values linearly chosen in the interval $[0, \pi]$.

Explanation: This exercise introduces element-wise vector operations in Scilab.

- For the first task, adding a scalar to a vector adds that scalar to every element identically: $x + 1 \Rightarrow [2, 3, 4, 5]$.
- The second task requires element-by-element multiplication (the Hadamard product). In Scilab, this is executed using the `.*` operator, multiplying elements at corresponding indices: $x .* y \Rightarrow [1 \times 5, 2 \times 6, 3 \times 7, 4 \times 8]$.
- The third task utilizes the `linspace(0, %pi, 10)` function to generate an array of $N = 10$ linearly spaced points. The `sin()` function is then mapped across the generated vector space.

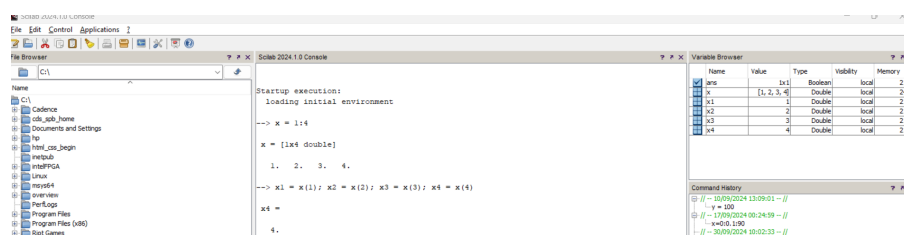


Figure 1: Result of Ex 1.1 - Part 1

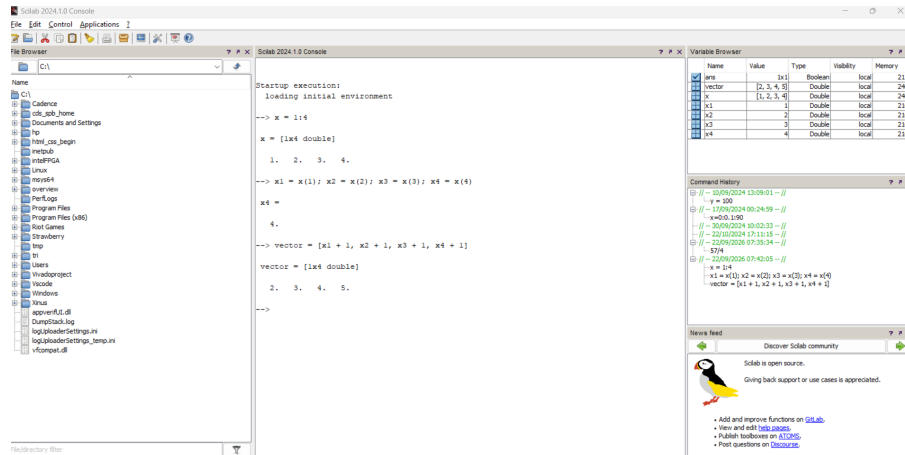


Figure 2: Result of Ex 1.1 - Part 2

Exercise 1.2

Question: Consider the analog signal $x_a(t) = 3 \sin(100\pi t)$.

- Use Scilab to draw $x_a(t)$ in 5 periods.
- Determine the discrete-time signal $x(n)$ of the signal $x_a(t)$ that is sampled with a sampling rate $F_s = 300$ samples/s.
- Determine the discrete-time signal $x(n)$ and determine its periodic property. If $x(n)$ is a periodic signal, determine the frequency and period of $x(n)$. Then, use Scilab to draw $x(n)$ in 5 periods.
- Determine the quantized signal $x_q(n)$ if $\Delta = 0.1$ using the truncated method. Then, draw $x_q(n)$ in 5 periods. All figures must be displayed in a single window.

Explanation: Given $x_a(t) = 3 \sin(100\pi t)$, the analog frequency is $F_0 = 50$ Hz. By applying the sampling theorem with $F_s = 300$ Hz, the sampling interval is $T_s = 1/F_s$. Substituting $t = nT_s$:

$$x(n) = x_a(nT_s) = 3 \sin\left(100\pi \frac{n}{300}\right) = 3 \sin\left(\frac{\pi}{3}n\right) \quad (1)$$

A discrete sinusoid is periodic if its digital frequency $f = \frac{F_0}{F_s} = \frac{50}{300} = \frac{1}{6}$ is a rational number. Because $f = \frac{1}{6}$, $x(n)$ is periodic. The fundamental period is determined by $N = \frac{K}{f}$ where K is the smallest integer yielding an integer N . Thus, for $K = 1$, the discrete period is $N = 6$ samples.

For quantization, we apply a truncation method with step size $\Delta = 0.1$. The truncated value $x_q(n)$ is obtained via the floor operation after normalizing by Δ :

$$x_q(n) = \Delta \left\lfloor \frac{x(n)}{\Delta} \right\rfloor = 0.1 \lfloor 10 \cdot x(n) \rfloor \quad (2)$$

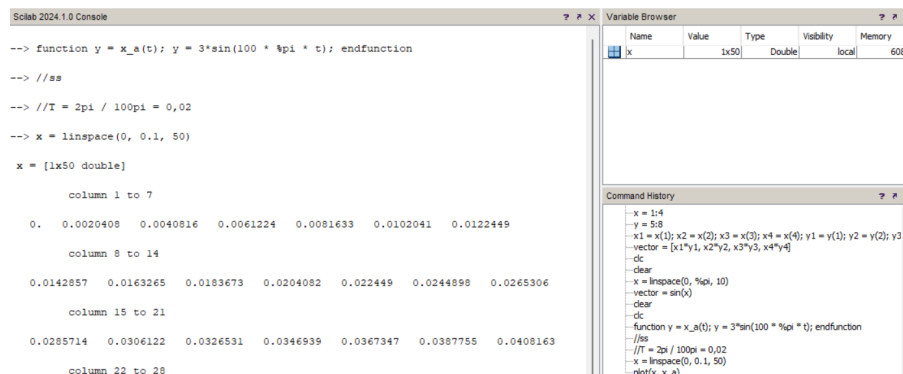


Figure 3: Result of Ex 1.2 - Analog and Discrete Signal

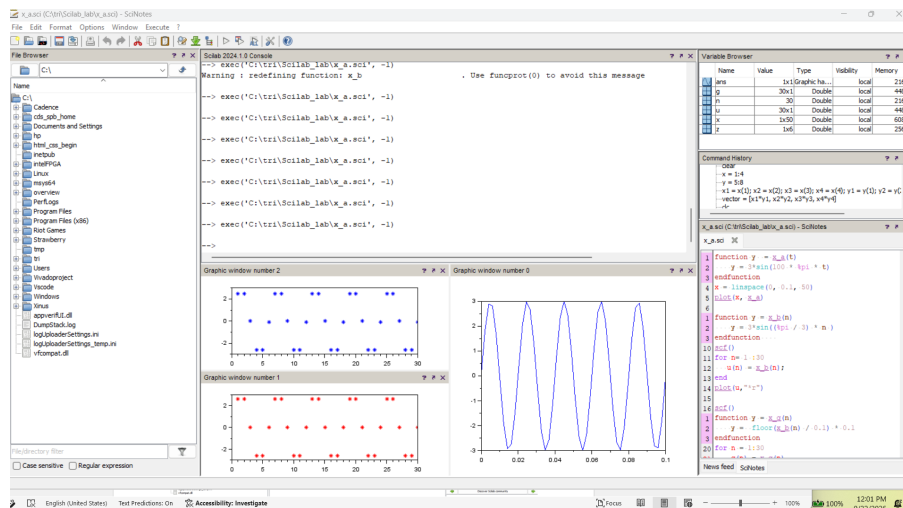


Figure 4: Result of Ex 1.2 - Quantized Signal and Errors

4.2 Lab 2

Exercise 1: Investigate Basic Functions

Question: Let's investigate the following functions on Scilab and briefly report their functionalities and how to use it: `plot2d3`, `min`, `max`, `subplot`, `title`, `xlabel`, `ylabel`, `bool2s`, `deff`.

Explanation: This exercise evaluates fundamental plotting and computational functions:

- `plot2d3(x, y)`: Plots vertical lines (stems) representing discrete-time samples. It is strictly used for discrete signals to highlight values at integer indices.
- `max()` and `min()`: Calculates the maximum/minimum elements in a matrix or vector array, highly useful for determining normalization constants or quantization step limits.
- `subplot(m, n, p)`: Divides the graphic window into an $m \times n$ matrix of sub-windows, selecting the p^{th} window for the active plot.

- `title()`, `xlabel()`, `ylabel()`: These functions add textual annotations to the graph. `title()` sets the main heading, while `xlabel()` and `ylabel()` label the respective axes, which is crucial for identifying discrete components properly.
- `bool2s()`: Converts a boolean array containing %T and %F into a numerical array of 1s and 0s. This is mathematically equivalent to the Heaviside step function evaluating a logic condition.
- `deff()`: Defines inline custom functions on the fly, similar to lambda expressions. Syntax: `deff("[y] = f(x)", "y = ...")`.

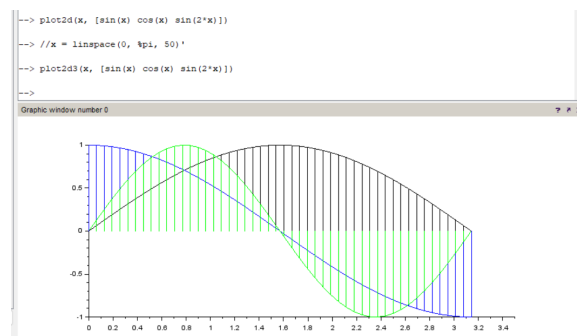


Figure 5: plot2d3 Usage

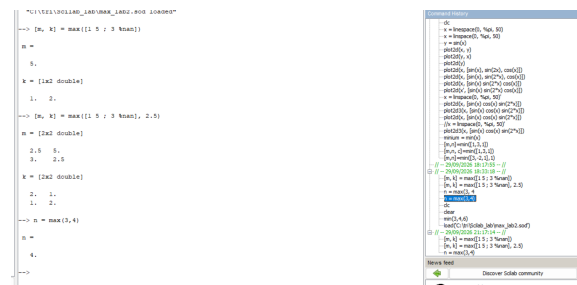


Figure 6: max() Usage

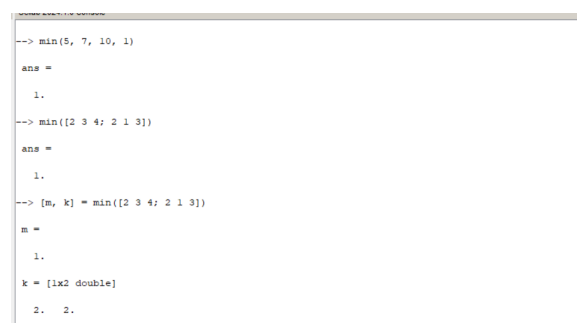


Figure 7: min() Usage

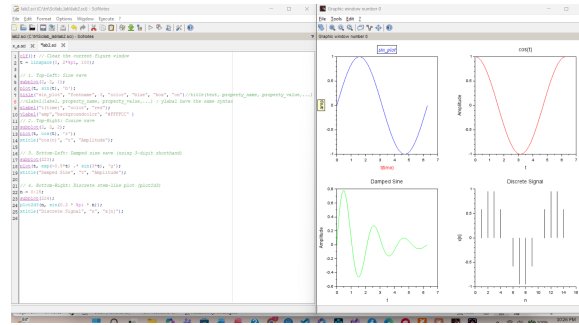


Figure 8: subplot() + title() + xlabel() + ylabel() Usage

```
--> deff("y = f(x)", " y = 3*x + 1");

--> f(5)

ans =

    16.
```

Figure 9: deff Usage

Exercise 2 & Exercise 3

Question: Try the following scripts on Scilab and report your understanding after observing the output.

- `n = -5:5; msignal = bool2s(n >= 0); plot2d3(n, msignal)`
- `n = -5:5; msignal = bool2s(n == 0); plot2d3(n, msignal)`

Explanation:

- **Exercise 2:** The logical condition `n >= 0` evaluates to True for $n \geq 0$ and False otherwise. `bool2s` transforms these booleans into numeric 1s and 0s. The output precisely replicates the mathematical definition of a **unit step sequence** $u(n)$:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (3)$$

- **Exercise 3:** The equality `n == 0` is true only at the origin point $n = 0$. Consequently, applying `bool2s` creates a sequence that represents the **unit sample sequence (discrete Dirac delta function)** $\delta(n)$:

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (4)$$

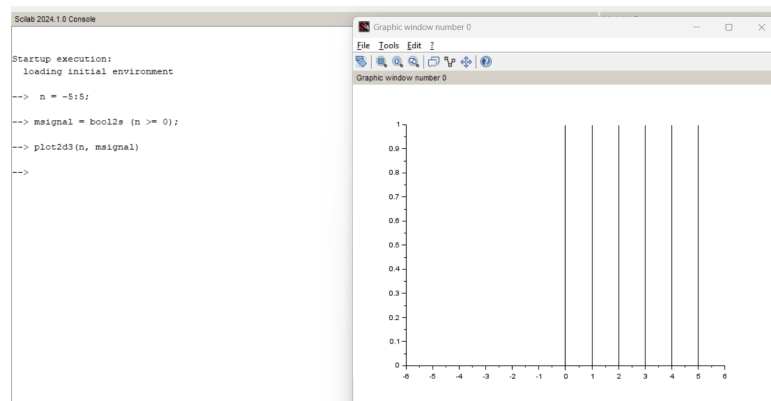


Figure 10: Result of Ex 2 - Unit Step Sequence $u(n)$

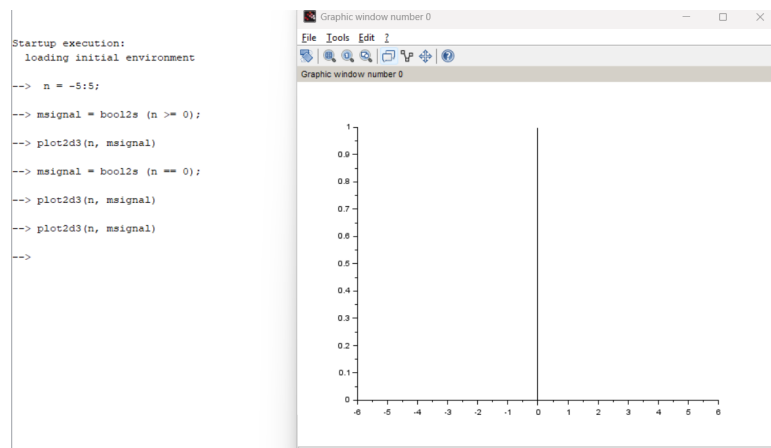


Figure 11: Result of Ex 3 - Unit Sample Sequence $\delta(n)$

Exercise 4: Unit Ramp Signal

Question: Use Scilab to draw the unit ramp signal $u_r(n)$ for $n = -5 : 5$.

Explanation: The discrete unit ramp signal is mathematically defined as the integral of the unit step function, or more precisely:

$$u_r(n) = n \cdot u(n) = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (5)$$

In Scilab, this is implemented natively by executing an element-wise product between the index space array n and the previously defined unit step array.

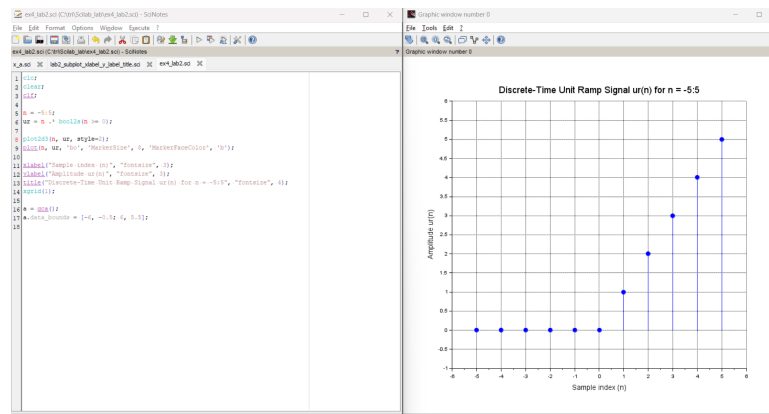


Figure 12: Result of Ex 4 - Unit Ramp Signal

Exercise 5: Even and Odd Components

Question: Given a discrete-time signal $x(n] = \{1, 3 \uparrow, -2\}$. Use Scilab to draw the signal $x(n)$, the odd signal component $x_o(n)$, and the even signal component $x_e(n)$.

Explanation: The arrow signifies the origin index $n = 0$. The sequence is thus explicitly indexed as $x(-1) = 1, x(0) = 3, x(1) = -2$. According to signal properties, any arbitrary sequence can be perfectly decomposed into an even-symmetric component $x_e(n)$ and an odd-symmetric component $x_o(n)$. We first evaluate the folded (time-reversed) sequence $x(-n)$. Using the decomposition formula:

$$x_e(n) = \frac{x(n) + x(-n)}{2}, \quad x_o(n) = \frac{x(n) - x(-n)}{2} \quad (6)$$

Calculations yield $x_e(-1) = -0.5, x_e(0) = 3, x_e(1) = -0.5$, and correspondingly $x_o(-1) = 1.5, x_o(0) = 0, x_o(1) = -1.5$. Scilab strictly mirrors these mathematical operations linearly. Therefore, the resulting sequences are:

$$x_e(n) = \{-0.5, 3 \uparrow, -0.5\}, \quad x_o(n) = \{1.5, 0 \uparrow, -1.5\} \quad (7)$$

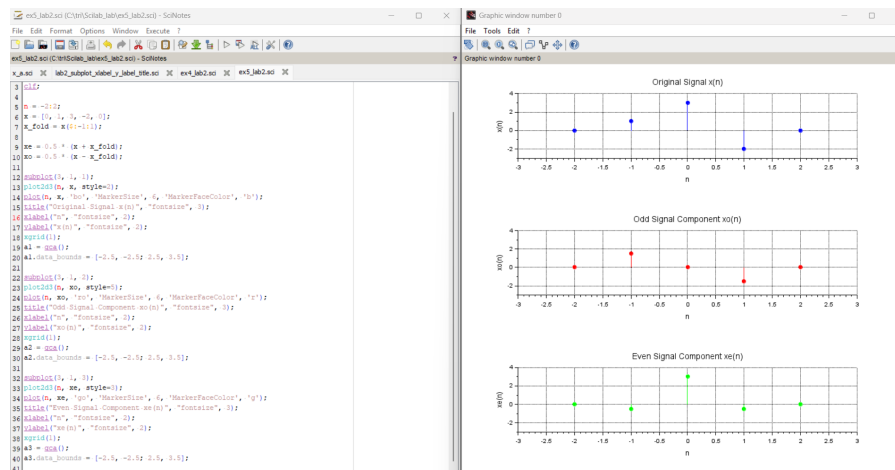


Figure 13: Result of Ex 5 - Even and Odd Decomposition

Exercise 6 & 7: Addition and Multiplication of Signals

Question:

- **Exercise 6:** Given two discrete-time signals $x_1(n) = \{0 \uparrow, 1, 3, -2\}$ and $x_2(n) = \{0, 1 \uparrow, 2, 3\}$. Determine $y(n) = x_1(n) + x_2(n)$.
- **Exercise 7:** Given the same signals, determine $y(n) = x_1(n) \cdot x_2(n)$.

Explanation: Because the two sequences span different bounded domain intervals ($x_1(n)$ spans $[0, 3]$ and $x_2(n)$ spans $[-1, 2]$), they cannot be directly operated on using primitive arrays. To implement a rigorous sample-by-sample operation, we determine the union of both index domains: $[\min(n_{1_{min}}, n_{2_{min}}), \max(n_{1_{max}}, n_{2_{max}})] = [-1, 3]$. Both vectors are then symmetrically zero-padded to conform to this unified domain grid. Once the sequences structurally align relative to the time origin $n = 0$:

$$y_{add}(n) = x_1(n) + x_2(n) = \{0, 1 \uparrow, 3, 6, -2\} \quad (8)$$

$$y_{mult}(n) = x_1(n) \cdot x_2(n) = \{0, 0 \uparrow, 2, 9, 0\} \text{ or simply } \{0 \uparrow, 2, 9\} \quad (9)$$

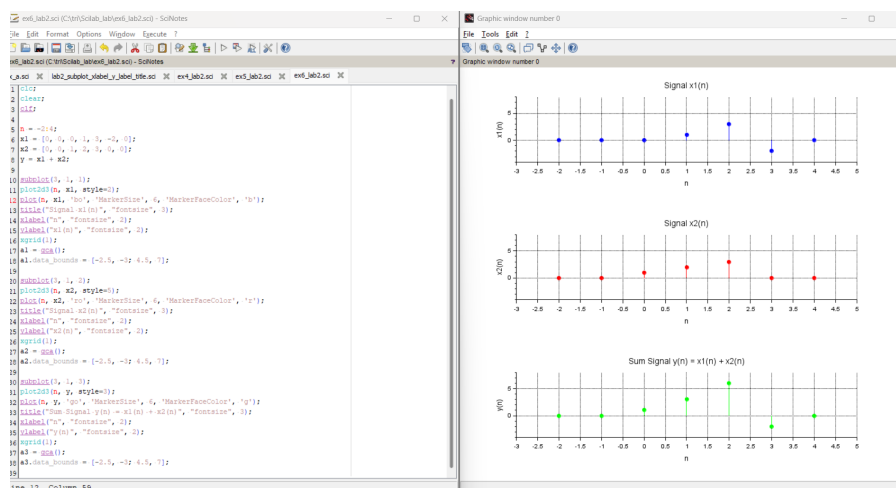


Figure 14: Result of Ex 6 - Addition of Signals

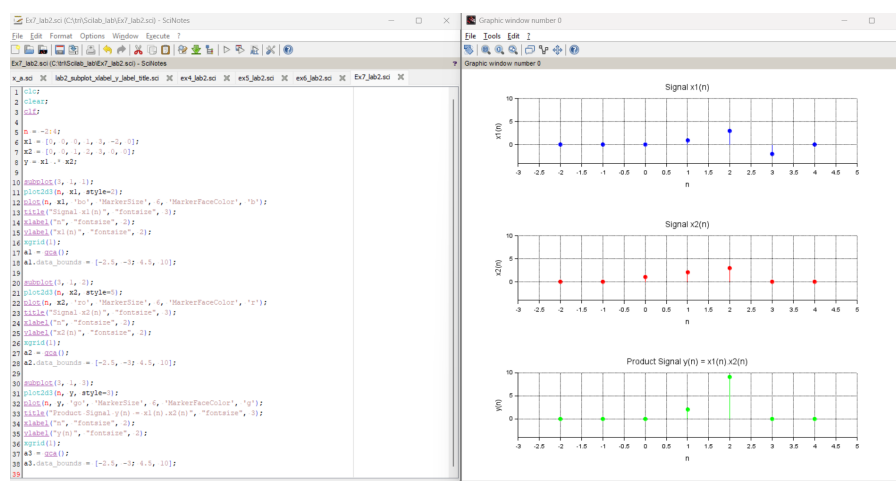


Figure 15: Result of Ex 7 - Multiplication of Signals

Exercise 8: Signal Manipulations

Question: Given a discrete-time signal $x(n] = \{1, -2, 3 \uparrow, 6\}$. Determine the following signals and draw both original and manipulated signals: 1. $y_1(n) = x(-n)$ 2. $y_2(n) = x(n + 3)$ 3. $y_3(n) = 2x(-n - 2)$

Explanation: For $x(n)$, the origin translates to $x(-2) = 1, x(-1) = -2, x(0) = 3, x(1) = 6$. We systematically apply independent operations on the sequence index variables:

1. **Folding (Time-Reversal):** $y_1(n) = x(-n)$. Every index evaluates to its symmetrical negative counterpart. Resulting indices: $[-1, 2]$.

$$y_1(n) = \{6, 3 \uparrow, -2, 1\} \quad (10)$$

2. **Advance (Time-Shift):** $y_2(n) = x(n + 3)$. The addition inside the argument denotes an advancement. The whole sequence is shifted to the left along the time axis by 3 sample units.

$$y_2(n) = \{1, -2, 3, 6, 0, 0 \uparrow\} \quad (11)$$

3. **Composite Operations:** $y_3(n) = 2x(-(n + 2))$. To evaluate this correctly: first perform folding to get $x(-n)$, then shift left by 2 to get $x(-(n + 2))$, and finally apply an amplitude scaling factor of 2.

$$y_3(n) = \{12, 6, -4, 2 \uparrow\} \quad (12)$$

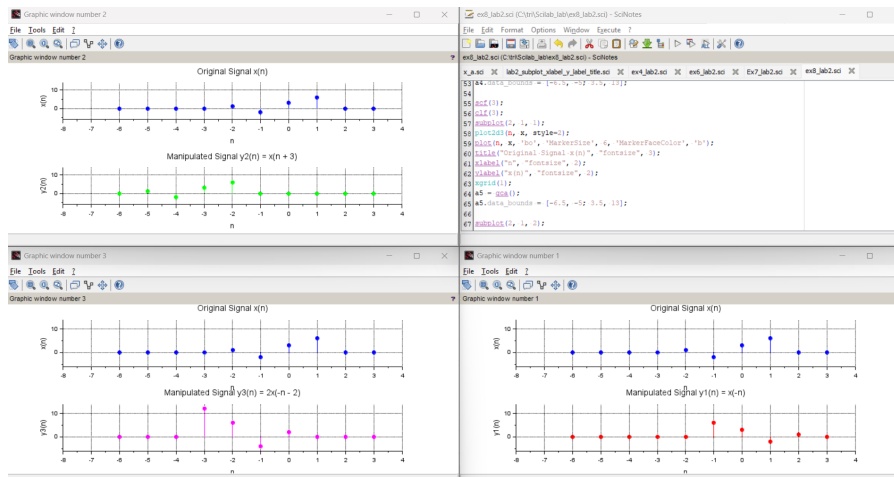


Figure 16: Result of Ex 8 - Signal Manipulations

Additional exercise 1.1

Additional exercise 1.3

Additional exercise 1.6

Additional exercise 1.7

Additional exercise 1.9

Additional exercise 1.10

5 Discussion

The laboratory thoroughly consolidated the theoretical understanding of discrete-time signals using systematic computational approaches. Scilab proved to be a versatile platform, allowing direct vector mapping of discrete mathematical formulas and offering visual validations through its `plot2d3` subroutines. Specifically, manual implementations of zero-padding demonstrated exactly how sequences must be mutually aligned relative to an origin reference prior to algebraic operations.

6 References

- Digital Signal Processing Lecture Notes - Ho Chi Minh City University of Technology.
- Scilab Online Manuals.